

Kunal Joshi

Cloud Security Architect and Engineer, Developer, Identity and Cryptography Subject Matter Expert

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SKILLS

Development

Java, Python, C/C++

Identity and Access Management

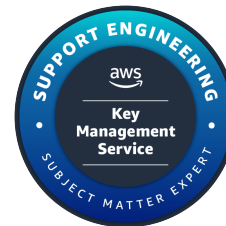
OAuth, SAML, AWS Identity And Access Management.

Cryptography

AWS Key Management Service, Hardware Security Modules (HSM), PKCS#11, JCE, OpenSSL Dyn. Encryption SDK.

Others

Penetration Testing, Privilege Escalation, Application Layer network design, Protection against DDoS, Protection against Owasp top 10 Web Application security risks, CloudFormation, AWS CDK, AWS SDKs, Terraform.



CERTIFICATIONS AND BADGES

AWS certified Security Specialty, AWS IAM Subject Matter Expert, AWS CloudHSM Subject Matter Expert, AWS KMS Subject Matter Expert.

EXPERIENCE

Amazon Web Services, Mumbai — Cloud Security Architect

May 2024 – Present

- Designed and implemented comprehensive security solutions for AWS customers to align with cloud security principles and regulatory compliance, focusing on Indian standards and regulations regulatory bodies in financial sector.
- Developed security architectures and action plans to adhere to security pillars including Identity and Access Management, Logging and Monitoring, Incident Response, Data Protection, and Infrastructure Security.
- Developed and deployed scalable security solutions, including automated least privilege policy creation, automated patch management systems, Infrastructure as Code (IaC) pipelines, and security scanners, enhancing the security posture for multiple customer environments.
- Worked closely with customers to translate complex security requirements into actionable cloud-based architectures, ensuring seamless integration with AWS services such as IAM, KMS, CloudTrail, and GuardDuty.

ACHIEVEMENTS

- ACM ICPC 2016-Kolkata finals **Rank 58.**
- Qualified - Google code jam 2017-2018
- #2223 - Round 1 Code Jam 2017
- AIR 98 - Tata Codevita 2016

Amazon Web Services, Bangalore — Cloud Engineer II - Security

May 2024 – Present

Cloud Security

- Managed escalations and issues with Security Domain AWS services such as IAM, KMS, CloudTrail, CloudHSM, Shield/DDoS, Cognito, Single-Sign On, GuardDuty, Inspector, AWS Organizations, Certificate Manager, etc.
- Developed internal training materials for services including Identity and Access Management, CloudHSM, Single Sign-On, and Secrets Manager

Subject Matter Expert - Identity and Access Management

- Served as a subject matter expert in AWS IAM services, specializing in concepts such as Federation, SAML, and Single Sign-On.
- Published public knowledge center articles such as - <https://aws.amazon.com/premiumsupport/knowledge-center/iam-tag-based-restriction-policies/>

AWS Single Sign On - Application Catalog team

- Contributed to onboarding **60+ applications to production** for [AWS Single Sign On - Cloud Application catalog](#) and authored public documentation for onboarded applications.

AWS SDK Operations

- Part of the AWS SDK Operations team, responsible for troubleshooting and debugging issues related to all AWS SDKs. Worked on AWS SDKs in multiple languages, including Python, Java, C++, C#, JavaScript, Ruby, Go, and PHP. Also managed other AWS tools such as AWS CLI, AWS CDK, and AWS Encryption SDK

Subject Matter Expert – Cryptography

- Subject Matter Expert in Cryptography services, specializing in AWS KMS and AWS CloudHSM.
- Handled escalations related to AWS KMS and CloudHSM issues.
- Worked on CloudHSM client using interfaces such as Java JCE, PKCS#11, and OpenSSL dynamic engine for backend cryptography operations.

OPEN-SOURCE PROJECTS

SpotLightPy - PyPI

- An Open Source Python Package (MIT License) published at Python Packaging Index (PyPI) to extract resource images from the Microsoft Spotlight Project for Windows X.
- Developed using Python 3.5
- Uses **Python Imaging Library (PIL)** to decrypt cached Images.
- **600+ downloads** by the global python community.
- **Git:** <https://github.com/joshikunal94/spotlightpy>
- **Python Packaging Index:** <https://pypi.python.org/pypi/spotlightpy>

EDUCATION

College Of Technology And Engineering, Udaipur — B.Tech Computer Science

2013 - 2017

Completed B.Tech from College Of Technology And Engineering, Udaipur with **O.G.P.A (Overall Grade Point) 7.98** with S.G.P.A of **8.49** in final semester.

Maharana Mewar Public School, Udaipur — Class XII

2012

Completed class XII examination of Central Board Of Secondary Education (CBSE) with **92.0%**.